Market Making on a New FX Exchange

Project Summary — EUR/USD Spot

Paris Dauphine University — MSc 203

April 2026

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1 Context and Architecture

We are the Designated Market Maker (DMM) on a new exchange A for EUR/USD spot. Two reference exchanges exist: exchange B (75% of total volume) and exchange C (25% of total volume), where price discovery occurs. Our structural constraint is a latency disadvantage: we observe B with a 200 ms delay and C with a 170 ms delay, whereas HFT firms access both in 50 ms, giving them a $\Delta\tau = 150$ ms advantage over us.

The project is divided into three one-month phases:

- **Phase 1** — Bootstrapping liquidity with no historical data (heuristics only).
- **Phase 2** — Calibrated market making after one month of data collection.
- **Phase 3** — Fallback liquidity provision once HFT market makers enter exchange A.

At each time step the algorithm: (1) observes the order books of B and C (with latency), (2) computes a reference mid-price, (3) computes optimal quotes, (4) submits and cancels orders on A, and (5) manages inventory and P&L.

2 Simulation of Exchanges B and C

2.1 Mid-price Model

Following the architecture validated with the course instructor, we adopt a **common exogenous mid-price with independent per-exchange noise** (Approach 1). A single fundamental mid-price S_t^* is simulated as a jump-diffusion process:

$$dS_t^* = \sigma_t \, dW_t + J_t \, dN_t, \tag{1}$$

where no drift term is included ($\mu = 0$) since EUR/USD exhibits no detectable drift at intraday horizons.

Intraday seasonality. Following [3], the instantaneous volatility incorporates a deterministic intraday seasonal component:

$$\sigma_t = \bar{\sigma} \cdot \varphi(t), \qquad \bar{\sigma} \approx 4\text{–}6 \text{ pips/hour}, \tag{2}$$

where $\varphi(t)$ is a piecewise function reflecting trading session activity:

Session	UTC hours	$\varphi(t)$
Tokyo	00h–08h	0.6
London open	08h–09h	1.4
London mid	09h–13h	1.0
London–NY overlap	13h–16h	1.5
Post-London	16h–18h	0.8
Overnight	18h–00h	0.5

Macro jumps. The jump component $J_t \, dN_t$ models macro announcements (NFP, CPI, central bank decisions):

$$N_t \sim \text{Poisson}(\lambda_{\text{jump}}), \quad \lambda_{\text{jump}} \approx 3\text{–}5 \text{ events/day}, \quad J_t \sim \mathcal{N}(0, \sigma_J^2), \quad \sigma_J \approx 5\text{–}10 \text{ pips}. \tag{3}$$

Jumps are implemented as a burst of directional market orders that consume several levels of the order book simultaneously on both exchanges, so that the price impact is endogenously generated by the queue dynamics rather than imposed exogenously.

Per-exchange noise. Each exchange receives an independent idiosyncratic noise component around S_t^*:

$$S_t^i = S_t^* + \varepsilon_t^i, \qquad i \in \{B, C\}, \tag{4}$$

where ε_t^B and ε_t^C are independent AR(1) processes:

$$\varepsilon_{t+dt}^i = \rho_\varepsilon \, \varepsilon_t^i + \sigma_\varepsilon \, \eta_t^i, \qquad \eta_t^i \sim \mathcal{N}(0,1), \qquad \rho_\varepsilon \approx 0.9, \qquad \sigma_\varepsilon \approx 0.1\text{–}0.5 \text{ pips}. \tag{5}$$

This ensures that B and C never move identically tick-by-tick while remaining cointegrated around the same fundamental value. The noise amplitude is calibrated to remain well below the arbitrage cost threshold (≈ 5 pips), so no spurious arbitrage opportunities are created.

Remark 1. This architecture cleanly separates macro dynamics (controlled via S_t^*) from micro dynamics (realistic queue behaviour via the CST model). It was validated as the preferred approach by the course instructor: the alternative of fully endogenous mid-prices with OU cointegration requires an additional synthetic arbitrageur agent with its own parameters to calibrate, adding substantial complexity without improving the quality of the market-making algorithm on exchange A.

2.2 Synthetic Order Book on B and C

Around the simulated mid-price S_t^i, we reconstruct a synthetic order book for each exchange using the stochastic queue model of [2]. For each price level p and side, the standing quantity $q_t(p)$ evolves according to three competing event types:

$$dq_t(p) = dA_t(p) - dC_t(p) - dE_t(p), \tag{6}$$

where A_t, C_t, E_t denote cumulative limit order arrivals, cancellations, and executions by market orders respectively.

Limit order arrivals. New LOs arrive at level p at a Poisson rate that decreases exponentially with depth $\delta = |p - S_t^i|$:

$$\lambda^a(\delta) = \lambda_0^a \cdot e^{-\alpha\delta}. \tag{7}$$

LOs arriving closer to the mid-price are more frequent, reflecting the well-documented empirical shape of order book density [1].

Cancellations. Each resting LO is cancelled independently at rate θ, so the aggregate cancellation rate at level p is proportional to the queue size:

$$\text{cancellation rate at } p = \theta \cdot q_t(p). \tag{8}$$

Empirically on EUR/USD, approximately 40–60% of LOs are cancelled before execution [14].

Market order arrivals and executions. Market orders arrive at rate λ^{MO} (homogeneous in Phase 1, state-dependent in Phase 2) and consume the best available level first. A MO of size v executes $\min(v, q_t(p^*))$ units at the best level p^*; any remaining volume walks to the next level. When $q_t(p^*) = 0$ after an execution, the best quote moves by one tick and the mid-price updates accordingly.

Spread process. The bid-ask spread is modelled as a two-state Markov chain:

$$\Delta_t \in \{1 \text{ pip}, \ 2 \text{ pips}\}, \tag{9}$$

with transition probabilities calibrated so that $\Delta_t = 1$ pip holds approximately 80% of the time during liquid sessions, consistent with EUR/USD behaviour on EBS.

Stochastic volume factor. Rather than fixing volumes at a deterministic level, we multiply mean queue sizes by a global liquidity factor η_t that follows a mean-reverting AR(1) process:

$$\eta_{t+dt} = \bar{\eta} + \rho\,(\eta_t - \bar{\eta}) + \sigma_\eta\,\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1), \tag{10}$$

with $\rho \approx 0.95$ (highly persistent liquidity) and $\sigma_\eta \approx 0.2$ (20% fluctuations around the mean). The mean volume at depth level k is then:

$$V_k(t) = \mu_k \cdot \eta_t, \qquad \mu_k = V_0 \cdot \beta^{k-1}, \qquad \beta \approx 1.3, \tag{11}$$

where volumes increase with depth because deep resting orders accumulate over longer periods before being reached by market orders.

Discrete-time simulation. In practice, with time step dt (e.g. 10 ms), the update at each level proceeds as follows:

1. LO arrivals: $N_{\text{arr}} \sim \text{Poisson}(\lambda^a(k) \cdot dt)$; $\quad q_k \mathrel{+}= N_{\text{arr}} \cdot v_{\text{unit}}$.
2. Cancellations: $N_{\text{can}} \sim \text{Poisson}(\theta \cdot q_k \cdot dt)$; $\quad q_k \mathrel{-}= \min(N_{\text{can}} \cdot v_{\text{unit}},\, q_k)$.
3. MO execution (if a MO arrives this step): consume from best level outward until MO volume exhausted.
4. Mid-price update: if $q_{\text{best}} = 0$, best quote advances one tick.

Market order arrivals. In Phase 1, market order arrivals are modelled as a homogeneous Poisson process with intensity λ^{MO} calibrated to known EUR/USD activity levels. In Phase 2, we replace this with the state-dependent intensity of [7]:

$$\lambda^{MO,\pm}(Z_t, \Delta_t), \tag{12}$$

where $Z_t \in \{1, \ldots, n_Z\}$ is the volume imbalance regime defined in Section 4.

2.3 Coherence Between CST Parameters and the Exogenous Mid-price

A key requirement of Approach 1 is that the queue dynamics governed by the CST intensities $(\lambda^{MO}, \lambda_0^a, \alpha, \theta)$ be *consistent* with the volatility of the exogenous process S_t^*. If the parameters are chosen independently, a contradiction arises: the order book queues may empty at a rate that is incompatible with the frequency at which S_t^* itself moves.

Role of best-queue depletion. In our implementation, the mid-price on each exchange updates by one tick whenever the best queue is fully consumed by incoming market orders. The exogenous process S_t^* therefore acts as a target: after each depletion event, the mid-price is snapped back to S_t^*. For this mechanism to be internally consistent, the *frequency at which best queues deplete* must match the *frequency at which S_t^* itself moves by one tick.*

Target depletion frequency. The expected time between one-tick moves of a discretised Brownian motion with volatility $\bar{\sigma}$ and tick size Δ_{tick} is:

$$\mathbb{E}[\tau_{\text{mid}}] \approx \frac{\Delta_{\text{tick}}^2}{\bar{\sigma}^2}. \tag{13}$$

With $\bar{\sigma} = 5$ pips/hour $= 1.39 \times 10^{-4}$ per $\text{s}^{1/2}$ and $\Delta_{\text{tick}} = 10^{-4}$, this gives:

$$\mathbb{E}[\tau_{\text{mid}}] \approx \frac{(10^{-4})^2}{(1.39 \times 10^{-4})^2} \approx 52 \text{ seconds}. \tag{14}$$

Under intraday seasonality, this target scales as $\varphi(t)^{-2}$: during the London–New York overlap ($\varphi = 1.5$), mid-price moves occur roughly every $52/2.25 \approx 23$ s, compared with once every $52/0.25 \approx 208$ s overnight.

Implied constraint on CST intensities. The mean stationary queue size at the best level is:

$$\bar{q}_{\text{best}} \approx \frac{\lambda_0^a}{\theta}. \tag{15}$$

Assuming unit-size market orders, the best queue is fully consumed after approximately $\bar{q}_{\text{best}}$ market order arrivals, giving a mean depletion time:

$$\mathbb{E}[\tau_{\text{dep}}] \approx \frac{\bar{q}_{\text{best}}}{\lambda^{MO}} = \frac{\lambda_0^a}{\theta \cdot \lambda^{MO}}. \tag{16}$$

The coherence condition $\mathbb{E}[\tau_{\text{dep}}] = \mathbb{E}[\tau_{\text{mid}}]$ yields:

$$\boxed{\frac{\lambda_0^a}{\theta \cdot \lambda^{MO}} \approx \frac{\Delta_{\text{tick}}^2}{\bar{\sigma}^2 \cdot \varphi(t)^2}.} \tag{17}$$

This is not a free parameter: given λ^{MO} calibrated from trade-count data [14] and $\bar{\sigma}$ from [3], equation (17) constrains the ratio λ_0^a/θ, i.e. the mean stationary depth at the best level.

Numerical check. Using $\lambda^{MO} \approx 0.5$ trades/s [14] and $\mathbb{E}[\tau_{\text{mid}}] \approx 52$ s during the liquid London session, equation (17) requires:

$$\bar{q}_{\text{best}} = \lambda^{MO} \cdot \mathbb{E}[\tau_{\text{mid}}] \approx 0.5 \times 52 \approx 26 \text{ units}, \tag{18}$$

which is compatible with typical EUR/USD best-queue depths on EBS and provides a self-consistent starting point for the simulation.

Verification procedure. Because no closed-form expression links CST intensities to the emergent depletion frequency in general — [2] computes depletion probabilities via Laplace transforms rather than simple closed-form ratios — we verify coherence *numerically*: we simulate the CST order book in isolation, measure the empirical mean inter-depletion time $\hat{\tau}_{\text{dep}}$, and iterate on λ_0^a and θ until $\hat{\tau}_{\text{dep}} \approx \mathbb{E}[\tau_{\text{mid}}]$ for each intraday session separately (to respect the seasonality scaling $\varphi(t)^2$). This simulation-based method of moments is the standard calibration approach used in [2].

3 Phase 1 — Heuristic Market Making

3.1 Reference Mid-price

We combine the (stale) last observed prices from B and C using a volume-weighted average:

$$S_t^{\text{ref}} = 0.75\, \hat{S}_t^B + 0.25\, \hat{S}_t^C, \tag{19}$$

where $\hat{S}_t^B$ is the last observed price on B, stale by 200 ms.

3.2 Reservation Price

Following [4], we quote around the *reservation price*, which adjusts for inventory risk:

$$r_t = S_t^{\text{ref}} - q_t \cdot \gamma \cdot \sigma^2 \cdot (T - t), \tag{20}$$

where q_t is the current inventory in EUR, $\gamma > 0$ is the risk-aversion parameter, σ^2 is the mid-price variance, and $T - t$ is the remaining time in the trading window. When $q_t > 0$ (long EUR), we have $r_t < S_t^{\text{ref}}$, shifting the entire order book downward to encourage sells and discourage buys.

3.3 Optimal Spread

The quoted spread is decomposed into three independent components:

$$\psi^* = \underbrace{\gamma\sigma^2(T-t) + \frac{2}{\kappa}\ln\Big(1+\frac{\gamma}{\kappa}\Big)}_{\psi_{\text{AS}}} + \underbrace{2\,\bar{\sigma}\sqrt{\Delta\tau}}_{\psi_{\text{snipe}}} + \underbrace{\psi_{\text{fees}}}_{\psi_{\text{fees}}}, \tag{21}$$

where κ is the order-flow elasticity with respect to depth.

Inventory and competition component ψ_{AS}. This is the closed-form spread derived in [12], which generalises [4] and provides existence and uniqueness guarantees. It covers inventory risk and the cost of providing liquidity in a competitive environment.

Sniping protection component ψ_{snipe}. Following [5], our quotes are stale for $\Delta\tau = 150\,\text{ms}$ after any price movement on B or C. An HFT observing a jump J on B will profitably trade against us on A whenever $J > \psi^*/2$. The protection component is:

$$\psi_{\text{snipe}} = 2\,\sigma_{\Delta\tau} = 2\,\bar{\sigma}\sqrt{\Delta\tau}, \tag{22}$$

where $\sigma_{\Delta\tau}$ is the standard deviation of EUR/USD price changes over the latency window $\Delta\tau$.

Fee recovery component ψ_{fees}. We recover maker fees on both sides of a round-trip trade:

$$\psi_{\text{fees}} = 2 \times 0.01\% = 2 \text{ pips}. \tag{23}$$

3.4 Order Book Construction on Exchange A — 10 Levels per Side

Our order book on exchange A is entirely under our control. We must decide (i) at which prices to post, (ii) what quantities to offer at each level, and (iii) when to cancel and repost after market moves.

Price grid. Prices are placed at geometrically increasing distances from r_t:

$$\text{ask}_k = r_t + \frac{\psi^*}{2} + \sum_{i=1}^{k-1}\delta_i, \qquad \delta_i = \delta_1 \cdot \lfloor 1.4^{i-1} \rfloor, \qquad k = 1,\ldots,10, \tag{24}$$

and symmetrically for bid levels. Each level is rounded outward (away from mid) to the nearest pip to ensure the theoretical protection floor is never breached. We do not occupy every intermediate tick: levels 1–3 are consecutive (1-pip spacing) while levels 4–10 use increasing spacing, reducing message traffic without sacrificing depth coverage.

Levels	Distance from mid	Role
1–3	$\psi^*/2$ to +2 pips	Active profit zone
4–7	+2 to +10 pips	Intermediate protection
8–10	+10 to +50 pips	Safety net (large orders, Phase 3)

Quantities at each level. The total potential inventory if all bid levels are filled must not exceed the remaining delta headroom $q_{\max} - |q_t|$. Under this constraint, quantities are allocated geometrically:

$$q_k = (q_{\max} - |q_t|) \cdot \alpha^{k-1} \cdot \frac{1-\alpha}{1-\alpha^{10}}, \qquad \alpha \approx 0.7, \tag{25}$$

which guarantees $\sum_{k=1}^{10} q_k = q_{\max} - |q_t|$ exactly. More liquidity is offered near the mid-price, where execution is more likely and the spread gain exceeds the adverse-selection cost. In Phase 2, the allocation is re-calibrated from observed fill rates: levels that are rarely executed receive smaller allocations.

Quote update triggers. We maintain a tolerance band $\varepsilon \approx 0.5$ pip around the last-posted reservation price. A full cancel-and-repost cycle is triggered by any of three events:

1. **Mid-price move:** $|\hat{S}_t^B - S_{\text{last}}| > \varepsilon$ — the reference price has shifted enough to make current quotes stale.
2. **Fill event:** one of our LOs is executed, changing q_t and thus r_t.
3. **Inventory alert:** $|q_t|$ crosses the 70% threshold — we enter the alert regime and must skew the book more aggressively.

Below the threshold ε, the cost of re-quoting (latency, execution risk during order replacement) exceeds the benefit of marginally better prices.

Cancel-and-repost logic. At each update event:

1. Compute new r_t from equation (20).
2. For each level $k = 1, \ldots, 10$: compute new ask_k, bid_k, q_k.
3. Cancel any existing order whose price differs from the new target.
4. Submit new orders at the updated prices and quantities.

3.5 Inventory Management and Hedging

Three regimes govern quoting and hedging behaviour:

Regime	Condition	Action
Normal	$\lvert q_t\rvert < 70\%\, q_{\max}$	Symmetric book, standard skew via r_t
Alert	$70\% \le \lvert q_t\rvert < 90\%\, q_{\max}$	Aggressive skew; reduce size on aggravating side
Hedge	$\lvert q_t\rvert \ge 90\%\, q_{\max}$	Immediate MO hedge on B (taker fee 0.02%)

Exchange B is preferred for hedging (taker fee 0.02% vs. 0.03% on C).

4 Phase 2 — Calibrated Market Making

4.1 Volume Imbalance Signal

After one month of Phase 1 data, we calibrate the model of [7]. The volume imbalance is:

$$\rho_t = \frac{V_t^b - V_t^a}{V_t^b + V_t^a} \in [-1, 1], \tag{26}$$

where V_t^b and V_t^a are volumes at the best bid and ask on exchange A. This continuous signal is discretised into n_Z regimes: $Z_t \in \{1, \ldots, n_Z\}$, with $Z_t = 1$ denoting sell-heavy and $Z_t = n_Z$ denoting buy-heavy conditions.

4.2 State-Dependent Parameters

For each imbalance–spread state $J = (Z, \Delta)$, we estimate from data:

- $\lambda^{MO,\pm}(J)$ — market order arrival intensities conditional on state J,
- $\epsilon^{\pm}(J) = \mathbb{E}[\Delta S \mid MO^{\pm}, J]$ — expected mid-price change immediately after a market order.

The quantity $\epsilon^{\pm}(J)$ is the key adverse selection measure: if $\epsilon^{+}(J)$ is large and positive, a buy MO is likely to be followed by a price increase, so our posted ask LO is being filled against an informed or momentum-driven trader.

4.3 Optimal Posting Rule

The HJB equation of [7] yields binary posting controls:

$$\gamma^+(t,q,J) = 1 \iff \frac{\Delta}{2} - \epsilon^+(J) + \Sigma_1^+(t,q,J) > \Sigma_0^+(t,q,J), \quad q \neq Q, \tag{27}$$

$$\gamma^-(t,q,J) = 1 \iff \frac{\Delta}{2} - \epsilon^-(J) + \Sigma_1^-(t,q,J) > \Sigma_0^-(t,q,J), \quad q \neq Q, \tag{28}$$

where $\Sigma_\gamma^\pm(t,q,J)$ captures expected future inventory and liquidation costs. The key quantity $\frac{\Delta}{2} - \epsilon^\pm(J)$ is the net expected gain from having a limit order filled: the half-spread earned minus the immediate adverse price move.

4.4 Intensity Function Calibration

Rather than assuming an exponential fill intensity $\Lambda(\delta) = Ae^{-k\delta}$ as in [4], we calibrate Λ non-parametrically from fill data, following [11]. Under mild regularity conditions (continuity, log-concavity, bounded profit per trade), the optimal depth equations admit a unique solution for any specification of Λ, encapsulated in the auxiliary function:

$$\varphi(u) = \sup_{\delta \geq 0} \Lambda(\delta)\,(\delta + u), \tag{29}$$

which we compute numerically from the empirically calibrated Λ.

5 Phase 3 — Fallback Liquidity Provider

5.1 Change of Role

Once HFT market makers enter exchange A with tight spreads, we no longer compete at the best level. Our role shifts to that of a *fallback liquidity provider* in the sense of [10] and [6]: we absorb large orders and provide liquidity during stress events (volatility spikes, HFT server crashes, one-sided markets).

5.2 Adapted Quoting Strategy

Normal conditions. We withdraw from levels 1–2 (where HFTs dominate) and concentrate liquidity on levels 3–10. We do not compete on the tight spread.

Stress conditions. When realised volatility spikes or HFT quotes disappear, our deep levels become the only available liquidity. The wide spread at these levels is economically justified as a risk premium for absorbing large imbalanced flows, consistent with the liquidity spiral mechanism of [6].

5.3 Dynamic Spread Adjustment

The spread adapts in real time to realised volatility:

$$\psi_3^*(t) = \psi_2^* \cdot \frac{\sigma_t^{\text{realized}}}{\bar{\sigma}}, \tag{30}$$

where $\sigma_t^{\text{realized}}$ is estimated over a rolling short window. When volatility doubles, our spread automatically doubles, providing protection during the precise moments when adverse selection and inventory risk are highest.

6 Summary

	Phase 1	Phase 2	Phase 3
Mid-price source	Weighted avg B/C	Weighted avg + OFI	Idem + vol adj.
Spread model	A-S + BCS + fees	CDJ calibrated	CDJ + dyn. vol
Active levels	1–3	1–3	3–10 only
Parameters	Heuristics	Calibrated from A	Re-calibrated
Hedge trigger	> 90% delta	Idem + OFI	Idem
Key references	[4, 5]	[7, 11]	[10, 6]

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